

□ CHAPTER 4: Project Design Phase

4.1 System Topology Pipeline Diagram

[User Interface] -> [Flask Core Controller] -> [Pre-Processing Logic] -> [Model Stack ML Inference] -> [Relational Sink Payload Output]

4.2 Database Schema Integrity Mapping

- • USER: Identity structures holding unique user codes, login contexts, and timestamps.
- • APPLICANT_PROFILE: Socio-demographic properties including gender matrices, marital status, dependent indexes, and geographic property area types.
- • CREDIT_HISTORY: Historical scoring tracks evaluating risk thresholds.
- • LOAN_APPLICATION: Monetary tracking elements detailing applicant base income, secondary coapplicant revenue, total loan values, and targeted repayment frequencies.
- • PREDICTION_RESULT: The runtime analytical sink compiling the specific prediction tracking metrics generated live by the web container backend.